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Lecturer: Paul McEvoy

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Proposal on -
Future Sales Prediction using Machine Learning and Deep
Learning Techniques

Submitted By
Dhanesh Joy Rodriguse, 10610107

Future Sales Prediction using Machine Learning and Deep Learning Techniques

ABSTRACT

Accurate sales forecasting is crucial for businesses to optimize inventory management, allocate resources efficiently, and improve customer satisfaction. This study compares the performance of advanced machine learning techniques, including attention mechanisms, graph neural networks (GNNs), recurrent neural networks (RNNs), long short-term memory (LSTM) networks, autoregressive integrated moving average (ARIMA), and vector autoregression (VAR) models, for sales forecasting using the "Predict Future Sales" dataset from Kaggle. The results demonstrate the effectiveness of these techniques in capturing complex temporal and relational patterns in sales data, providing valuable insights for businesses to make data-driven decisions.

1. INTRODUCTION

Sales forecasting plays a critical role in the success of a business. By accurately predicting future sales, businesses can manage inventory, allocate resources, and plan marketing strategies more effectively. Traditional sales forecasting methods, such as time series analysis and regression, have been widely used but may struggle with complex, high-dimensional datasets. The advent of advanced machine learning techniques, such as attention mechanisms and graph neural networks, has opened new avenues for improving sales forecasting. This study aims to investigate the performance of these techniques on the "Predict Future Sales" dataset from Kaggle, providing insights into their potential advantages and limitations.

2. LITERATURE REVIEW

1. Introduction

The rapid advancement of technology and the growth of e-commerce have led to an explosion in the amount of data generated by businesses. Companies can leverage this data to gain insights and make data-driven decisions that improve their sales and operational efficiency. One area of particular interest is sales forecasting, which

involves predicting future sales based on historical data and other factors. Accurate sales forecasts enable businesses to optimize inventory management, allocate resources efficiently, and improve customer satisfaction.

Machine learning (ML) and deep learning (DL) techniques have emerged as powerful tools for addressing complex forecasting problems in various domains, including sales forecasting. This literature review aims to explore the latest research in applying ML and DL techniques to future sales prediction and to identify the most promising methods and potential improvements.

2. Sentiment Analysis Techniques

2.2.1 Traditional Sales Forecasting Methods

Traditional sales forecasting methods, such as time series analysis, have long been used to predict future sales (Hyndman & Athanasopoulos, 2018). These methods usually involve analyzing historical sales data and identifying patterns, trends, and seasonality. Although these methods have been successful in many cases, they often struggle with non-linear relationships and large, high-dimensional datasets (Wang, 2021).

2.2.2 Machine Learning Techniques for Sales Forecasting

Machine learning-based sentiment analysis techniques employ supervised, unsupervised, or semi-supervised algorithms to classify text sentiment (Pang et al., 2008). Commonly used algorithms include Naive Bayes (NB) (McCallum & Nigam, 1998), Support Vector Machines (SVM) (Joachims, 2002), and Decision Trees (DT) (Quinlan, 1986). Recent research has focused on deep learning techniques, such as Convolutional Neural Networks (CNN) (Kim, 2014) and Recurrent Neural Networks (RNN) (Tang et al., 2015). These techniques have demonstrated improved performance in various sentiment analysis tasks (Zhang et al., 2018).

2.2.3 Deep Learning Techniques for Sales Forecasting

Deep learning techniques, which involve the use of artificial neural networks with multiple hidden layers, have shown promise in handling complex forecasting problems. Convolutional neural networks (CNN) have been applied to sales forecasting, with success in capturing spatial and temporal patterns in data (Borovykh et al., 2017). Recurrent neural networks (RNN), specifically long short-term memory (LSTM) networks, have demonstrated effectiveness in capturing temporal dependencies in time series data (Hochreiter & Schmidhuber, 1997).

Recent research has focused on combining CNN and LSTM networks in a hybrid approach to address the limitations of each model. For example, Yang et al. (2018) proposed a model called ConvLSTM, which integrates convolutional layers within an LSTM framework. This model achieved state-of-the-art performance on multiple sales forecasting benchmarks.

2.2.4 Feature Engineering and Selection in Sales Forecasting

Feature engineering and selection play a critical role in improving the accuracy of sales forecasts. Various studies have explored different features, such as product attributes, customer demographics, and external factors like holidays and promotions (Agarwal et al., 2019). Dimensionality reduction techniques, such as principal component analysis (PCA) and t-distributed stochastic neighbor embedding (t-SNE), have been employed to reduce the number of features while preserving important information (Maaten & Hinton, 2008).

2.2.5 Evaluation Metrics and Benchmarking

A variety of evaluation metrics are used to assess the performance of sales forecasting models, including mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE) (Chai & Draxler, 2014). The choice of evaluation metric depends on the specific problem and the characteristics of the data. Recent studies have also proposed the use of pinball loss (Fasiolo et al., 2021) and symmetric mean absolute percentage error (SMAPE) (Makridakis, 1993) as additional evaluation metrics.

Benchmarking is essential for comparing the performance of different models and identifying the most promising techniques. Several datasets have been used for benchmarking sales forecasting models, such as the M3-Competition dataset (Makridakis & Hibon, 2000) and the UCI Online Retail dataset (Lichman, 2013). More recently, the Walmart Sales Forecasting dataset (Kaggle, 2014)

and the Rossman Store Sales dataset (Kaggle, 2015) have been used to evaluate the performance of ML and DL techniques in sales forecasting.

2.3 Conclusion

The literature review has highlighted the growing interest in applying ML and DL techniques to future sales prediction. While traditional time series methods remain relevant, recent advances in ML and DL have demonstrated potential improvements in accuracy and the ability to handle complex, high-dimensional datasets. Ensemble techniques, XGBoost, CNN, LSTM, and hybrid approaches have shown promise in sales forecasting. Furthermore, feature engineering and selection, as well as the choice of evaluation metric, play crucial roles in the performance of sales forecasting models.

Future research should focus on developing novel ML and DL techniques tailored to specific sales forecasting challenges and further investigating the impact of feature engineering and selection. In addition, it is crucial to establish a comprehensive benchmarking framework to facilitate the comparison of different models and the identification of the most promising approaches.

Research Question

The main research question of this study is: How do advanced machine learning techniques, including attention mechanisms, GNNs, RNNs, LSTM, ARIMA, and VAR, compare in terms of accuracy and performance for sales forecasting using the "Predict Future Sales" dataset?

3. METHODOLOGY

The dataset consists of historical sales data for 3,340 products across 60 stores <https://www.kaggle.com/competitions/competitive-data-science-predict-future-sales/data>. The data includes daily sales records, item prices, and item category information, spanning from January 2013 to October 2015.

Data Preprocessing

- Impute missing values and remove outliers
- Aggregate the daily sales data into monthly sales data
- Encode categorical variables (e.g., item categories)

Feature Engineering

- Extract temporal features (e.g., month, year, seasonality)
- Calculate lagged features (e.g., sales and item prices from previous months)
- Construct features for GNNs (e.g., product relationships, store relationships)

Model Training and Evaluation

- Split the dataset into training, validation, and test sets
- Train and tune the models (attention mechanisms, GNNs, RNNs, LSTM, ARIMA, VAR) using the training and validation sets
- Evaluate the models on the test set using performance metrics such as mean squared error (MSE) and mean absolute percentage error (MAPE)

Model Comparison

- Compare the performance of the models on the test set
- Analyze the advantages and limitations of each technique

4. ETHICAL ISSUES

Data privacy and security: The data used for sales prediction might contain sensitive information about customers or suppliers, such as personal details or financial data. It is crucial to protect the privacy of individuals by anonymizing data and following relevant data protection laws and regulations.

Bias and fairness: The datasets used for training machine learning and deep learning models might contain inherent biases, which can lead to unfair or discriminatory predictions. Researchers should carefully evaluate their data sources, models, and algorithms to minimize any unintended biases.

5. CONCLUSION

This study provides a comprehensive comparison of advanced machine learning techniques for sales forecasting using the "Predict Future Sales" dataset from Kaggle. By analyzing the performance of attention mechanisms, GNNs, RNNs, LSTM, ARIMA, and VAR models, the study offers valuable insights into the potential of these techniques in addressing complex sales forecasting problems. The findings will contribute to the development of more accurate and robust sales forecasting models, enabling businesses to make better-informed decisions and improve overall efficiency.

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